

OCVS target sizing

EXECUTIVE SUMMARY

The fictional estate sizes to **4 x BM.Standard.E5.192** hosts using the **192-core configuration**: three workload hosts plus one active N+1 reserve. The 231 powered-on VMs contribute 2,054 configured vCPUs and 6.48 TiB of configured memory. At the default 4:1 CPU ratio, the design reaches 89.1% of its planning envelope with one host unavailable and requires **768 portable VCF cores** based on full physical silicon.

231

Powered-on workloads

2,054

Configured vCPUs

6.48 TiB

Configured memory

4

Target hosts incl. N+1

768

Full-silicon VCF cores

89.1%

Failure-envelope occupancy

Assessment perimeter

Dimension	Synthetic value	Sizing treatment
Workloads	248 VMs + 8 templates	231 powered-on VMs size compute; 17 powered-off VMs are sensitivity only
Source estate	12 hosts / 3 clusters	10 AMD hosts and 2 Intel hosts
Powered-on demand	2,054 vCPU / 6.48 TiB memory	Configuration totals, not sustained utilization
Largest powered-on VM	32 vCPU / 1.25 TiB memory	Must fit on one target host
VM storage	180 TiB provisioned / 127 TiB in use	Designed separately from compute

Every identifier, quantity, ratio, and workload profile in this report is invented for demonstration. The values extend the same fictional estate used by the existing synthetic health-check and OCVS migration samples.

Recommended OCVS target

Design item	Recommendation	Why it matters
Shape	BM.Standard.E5.192 / 192-core configuration	AMD target aligned with most source hosts
Host count	3 workload + 1 N+1 = 4 total	Meets Oracle's three-host production minimum
Compute capacity	192 configured physical cores per host	Workload fit uses configured physical cores
VCF licensing	192 silicon cores per host / 768 total	Licensing uses every physical silicon core
Memory	2,304 GiB per host / 9 TiB total	The 1.25 TiB VM fits on one host
Storage	OCI Block Volume design required	Standard shapes do not use local dense vSAN storage

Sizing calculation

Step	Calculation	Result
Workload hosts	$\text{ceil}(2,054 \text{ vCPU} / (192 \text{ cores} \times 4.0))$	3
N+1 reserve	One additional active host	1
Total target hosts	3 workload + 1 reserve	4
Portable VCF cores	4 hosts x 192 physical silicon cores	768

Operating envelope

State	Available hosts	CPU ratio	4:1 envelope used	Memory ratio
Normal operation	4	2.67:1	66.9%	0.72:1
One host unavailable	3	3.57:1	89.1%	0.96:1

The N+1 host is active capacity, not a cold standby. During normal operation all four hosts may carry workloads. The design must still be able to run the estate on the three remaining hosts during failure or patching.

What the recommendation is sensitive to

HIGH - CPU EVIDENCE

Inventory is not performance history

The 4:1 ratio is a planning assumption. Validate several weeks of CPU demand, ready time, reservations, NUMA behavior, and latency-sensitive workloads before procurement.

HIGH - CPU VENDOR

The source mixes AMD and Intel

The recommended AMD cluster supports same-vendor live mobility for the AMD subset. The Intel subset needs a validated powered-off method or a separate Intel target cluster.

MEDIUM - LARGE VM

Two VMs exceed 1 TiB memory

The E5 host has enough memory for the invented 1.25 TiB maximum. Validate reservations, NUMA fit, active memory, and application support before assigning a migration method.

MEDIUM - POWERED OFF

Seventeen VMs are outside the baseline

Returning all invented powered-off demand adds 162 vCPUs and 0.42 TiB. The four-host design still fits at 4:1, but the decision needs owner confirmation.

MEDIUM - MANAGEMENT

No generic overhead was added

The default policy applies no management or growth uplift. Confirm the OCVS management footprint and leave operational headroom before final design approval.

OBSERVED - MINIMUM

The design meets the production minimum

Oracle documents a minimum of three hosts for a production multi-host SDDC. The recommendation has four total hosts and preserves one-host reserve.

Shape alternatives and the silicon-core effect

Shape	Hosts incl. N+1	Configured cores / host	Silicon cores / host	VCF cores	Decision
BM.Standard.E5.192 192-core config	4	192	192	768	Recommended; fits the 1.25 TiB VM
BM.Standard.E5.192 96-core config	7	96	192	1,344	Reduced compute does not reduce full-silicon licensing
BM.Standard.E4.128 128-core config	6	128	128	768	Fits large VMs, but needs more hosts
BM.Standard3.64 64-core config	10	64	64	640	Lower VCF count, but the 1.25 TiB VM cannot fit

The reduced-core E5 option is intentionally shown because it looks smaller from a compute perspective while requiring more VCF licenses in this example. Broadcom's guidance counts all physical CPU cores running the software and warns that disabled cores can make automated counts inaccurate.

Storage planning

Measure	Synthetic value	Use in design
VM data in use	127 TiB	Current consumed-data reference
Illustrative 30% operational floor	165.1 TiB	Operational slack, not compute growth
Provisioned virtual namespace	180 TiB	Must remain representable unless workloads are right-sized

Standard OCVS shapes use OCI Block Volume storage. Final capacity must cover performance, snapshots, backups, migration staging, service limits, and the required virtual namespace. VCF vSAN entitlement does not replace the OCVS Block Volume design.

Recommended next steps

- 01 Validate demand**
Collect several weeks of CPU, active memory, contention, storage, and network evidence. Re-run sizing if 4:1 is not supported.
- 02 Separate CPU-vendor paths**
Map workloads to AMD and Intel source hosts, then decide whether the Intel subset gets a separate target or a powered-off migration path.
- 03 Confirm large-VM fit**
Validate memory reservations, NUMA, application support, and migration method for the two VMs above 1 TiB.
- 04 Design Block Volume storage**
Model the 180 TiB namespace, performance, backup, staging, and operational headroom independently from compute.
- 05 Check the live catalog**
Confirm OCI region capacity, shape availability, VMware bundle, licensing, and commercial terms before commitment.

Assumptions and coverage gaps

	All values are fictional and cannot support an operational or commercial decision.
	The sizing baseline includes powered-on, non-template workloads only. Powered-off demand is reported separately.
	The 4:1 CPU ratio, 0% growth uplift, 0% generic management allowance, accepted aggregate memory overcommit, and one-host N+1 reserve are explicit planning assumptions.
	The recommendation uses full physical silicon for VCF licensing, even when a provider offers a reduced-core configuration.
	Reservations, limits, NUMA, CPU Ready, active memory, ballooning, swapping, storage latency and throughput, network demand, application dependencies, and business criticality are not available.
	OCI region, Availability Domain, capacity reservation, pricing, commitment term, VMware software bundle, and commercial agreement remain unselected.
	The target catalog and licensing rules were reviewed against public primary-vendor documentation on 11 August 2026.

Method and public sources

This report applies the RVTools Analyzer default sizing policy to fabricated inventory. Workload fit uses configured physical cores. Portable VCF licensing uses full physical silicon in every purchased host. Storage is assessed separately from compute. The report is a planning demonstration, not a quote or validated target design.

Source	How it is used
Oracle Cloud VMware Solution overview and supported shapes	Shape configurations, host minimums, processor-vendor rules, and standard-shape storage characteristics
Broadcom core-counting guidance	Physical-core licensing, 16-core minimum per CPU, and disabled-core warning

Acronym glossary

Term	Meaning	Term	Meaning
N+1	Workload capacity plus one host for failure or maintenance	OCVS	Oracle Cloud VMware Solution
SDDC	Software-defined data center	VCF	VMware Cloud Foundation
vCPU	Virtual CPU	VM	Virtual machine

Synthetic demonstration only. No customer data, source workbook, identifiers, hashes, or derived values were used.